

# EXAMINATIONS COUNCIL OF ZAMBIA



## Examination for General Certificate of Education Ordinary Level

### Biology

5090/1

### Paper 1 Multiple Choice

Thursday

8 AUGUST 2024

#### Additional Materials:

Multiple Choice Answer Sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Time: 50 minutes

Marks: 40

#### Instructions to Candidates

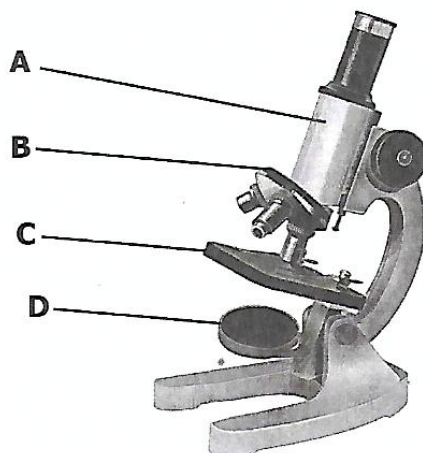
1. Ensure that the **school/centre name** and **subject paper, subject code, paper number, centre code, your examination number** and the **year** are correctly printed and shaded on the Answer Sheet.
2. There are **forty** questions in this paper. Answer **all** questions.
3. For each question there are four possible answers: **A, B, C and D**. Choose the **one** you consider correct and shade your choice using a **soft pencil** on the separate Answer Sheet.
4. Read **very carefully** the instructions on the Answer Sheet.

#### Information for Candidates

1. Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
2. Any rough working should be done in this Question Paper.
3. Cell phones and other electronic devices are **not allowed** in the examination room.



- 1 The diagram shows a light microscope.



Which labelled part supports the objective lens?

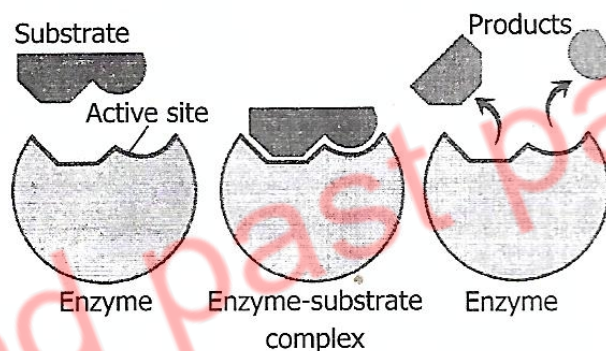
- 2 Which of the following describes the similarities between a plant cell and an animal cell?  
Both have ...

- A cell membrane and cell wall.
- B cell membrane and nucleus.
- C chloroplasts and vacuole.
- D cytoplasm and vacuole.

- 3 Which of the following statements correctly defines active transport? Movement of ...

- A molecules across a membrane along a concentration gradient using energy from ATP.
- B molecules across a membrane against a concentration gradient using energy from ATP.
- C solutes into and out of a cell across a membrane.
- D water molecules from high concentration to low concentration across a membrane.

- 4 The diagram illustrates one model of enzyme action.



What model of enzyme action is being demonstrated in the diagram? Enzymes are ...

- A affected by inhibitors.
- B affected by pH.
- C affected by temperature.
- D substrate specific.

- 5 The table shows types of food and their protein content per gram.

| FOOD   | PROTEIN CONTENT/G |
|--------|-------------------|
| Apple  | 0.2               |
| Beef   | 19.0              |
| Carrot | 0.7               |
| Cheese | 26.0              |
| Milk   | 3.3               |

Which type of food would give the highest positive result with biuret test?

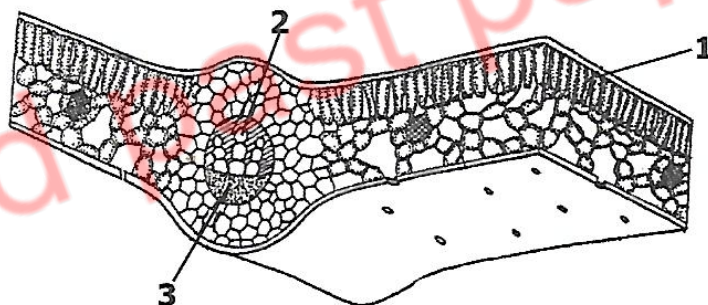
- A Apple and milk  
 B Beef and cheese  
 C Carrot and apple  
 D Milk and carrot
- 6 Which of the following diseases is as a result of iodine deficiency in the diet?  
 A Anaemia  
 B Goitre  
 C Marasmus  
 D Obesity

- 7 The table shows the recommended amounts of four nutrients for four different people.

| Person | Protein/g | Vitamin/g | Iron/g | Calcium/g |
|--------|-----------|-----------|--------|-----------|
| A      | 30        | 20        | 7      | 600       |
| B      | 43        | 20        | 10     | 600       |
| C      | 53        | 25        | 13     | 700       |
| D      | 60        | 60        | 14     | 1 200     |

Which person represents a pregnant woman?

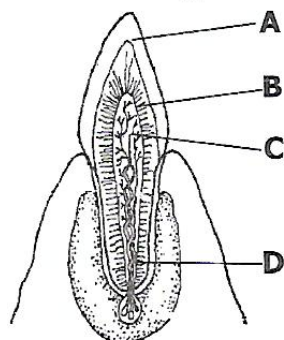
- 8 Which of the following shows the effect to a plant growing in a soil that lacks calcium?  
 A Poor fruit formation  
 B Poor root growth  
 C Stunted growth  
 D Yellowing of leaves
- 9 The diagram shows a transverse section of a dicotyledonous leaf.



What are the numbered parts?

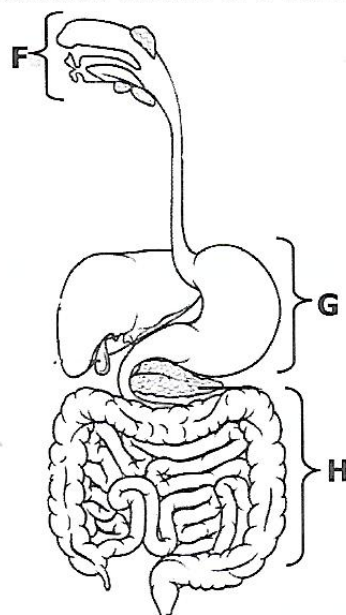
|   | 1        | 2      | 3      |
|---|----------|--------|--------|
| A | Cuticle  | Phloem | Xylem  |
| B | Cuticle  | Xylem  | Phloem |
| C | Palisade | Phloem | Xylem  |
| D | Palisade | Xylem  | Phloem |

- 10 The diagram shows the longitudinal section through a canine tooth.



Which part detects heat?

- 11 The diagram shows a section of a human alimentary canal?



Which events are correctly matched with the parts labelled F, G and H?

|   | F                          | G                          | H                          |
|---|----------------------------|----------------------------|----------------------------|
| A | Digestion of carbohydrates | Digestion of proteins      | Digestion of lipids        |
| B | Digestion of lipids        | Digestion of carbohydrates | Digestion of proteins      |
| C | Digestion of lipids        | Digestion of lipids        | Digestion of proteins      |
| D | Digestion of proteins      | Digestion of lipids        | Digestion of carbohydrates |

- 12 The following equation represents respiration in living organisms.

Glucose  $\longrightarrow$  X + carbon dioxide + energy (210KJ)

What is substance X in the equation?

- A Ethanol
- B Lactic acid
- C Oxygen
- D Water



13 Which of the following is the causative agent for malaria?

- ☒ A Female mosquito
- ☐ B Myobacteria
- ☐ C Plasmodium
- ☐ D Vibrio Cholerae

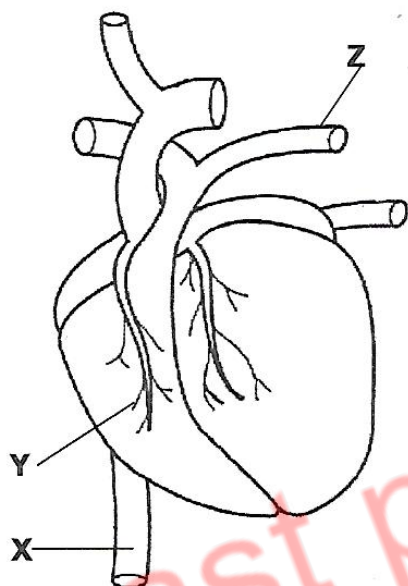
14 Which of the following is a safe sexual practice?

- ☒ A Consistent and correct use of condoms
- ☐ B Having multiple sexual partners
- ☐ C Having unprotected sexual intercourse
- ☐ D Kissing and sharing contaminated body fluids

15 Which of the following correctly shows the functions of the solid parts of blood?

|          | <b>Erythrocytes</b>       | <b>Leucocytes</b>         | <b>Thrombocytes</b>       |
|----------|---------------------------|---------------------------|---------------------------|
| <b>A</b> | Blood clotting            | Transport oxygen          | Defence against pathogens |
| <b>B</b> | Defence against pathogens | Transport oxygen          | Blood clotting            |
| <b>C</b> | Transport oxygen          | Blood clotting            | Defence against pathogens |
| <b>D</b> | Transport oxygen          | Defence against pathogens | Blood clotting            |

16 The diagram shows the external structure of the human heart.



Name the parts labelled X, Y and Z.

|          | <b>X</b>           | <b>Y</b>           | <b>Z</b>           |
|----------|--------------------|--------------------|--------------------|
| <b>A</b> | Aorta              | Coronary artery    | Superior vena cava |
| <b>B</b> | Aorta              | Superior vena cava | Pulmonary artery   |
| <b>C</b> | Inferior vena cava | Coronary artery    | Pulmonary artery   |
| <b>D</b> | Pulmonary artery   | Aorta              | Pulmonary vein     |

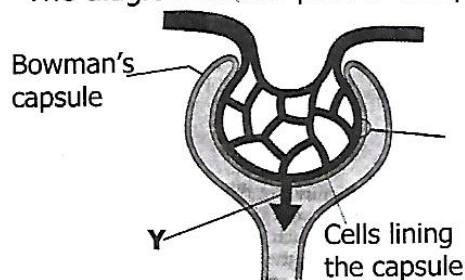
17 What is the difference between plasma and tissue fluid?

|   | Plasma                    | Tissue fluid              |
|---|---------------------------|---------------------------|
| A | Dissolved glucose present | No dissolved glucose      |
| B | Less dissolved glucose    | More dissolved glucose    |
| C | More protein molecules    | Fewer protein molecules   |
| D | No white blood cells      | White blood cells present |

18 Which of the following is **not** an excretory product of plants?

- A Carbon dioxide
- B Latex
- C Oxygen
- D Urea

19 The diagram shows part of a nephron.



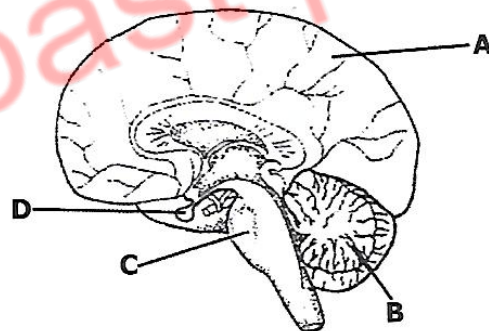
Which of the following is **not** found in the filtrate Y?

- A Glucose
- B Mineral ions
- C Proteins
- D Urea

20 Which of the following is **not** an endocrine gland?

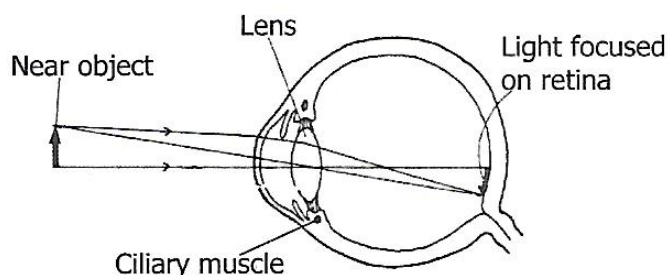
- A Hypothalamus
- B Pancreas
- C Pituitary
- D Thyroid

21 The diagram shows the human brain.



Which of the labelled parts is responsible for coordinating all body movements?

- 22 The diagram shows an eye accommodated for nearby objects.



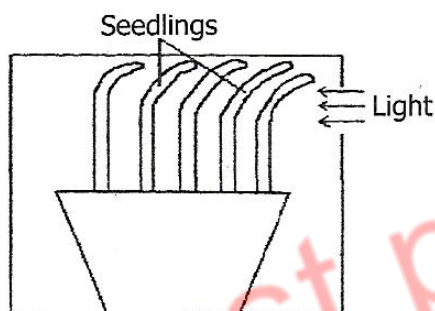
What happens to the ciliary muscle and lens?

|   | Ciliary muscle | Lens    |
|---|----------------|---------|
| A | Contracts      | Thicker |
| B | Contracts      | Thinner |
| C | Relaxes        | Thicker |
| D | Relaxes        | Thinner |

- 23 How do the biceps and triceps muscles work in straightening the arm?

|   | Biceps    | Triceps   |
|---|-----------|-----------|
| A | Contracts | Contracts |
| B | Contracts | Relaxes   |
| C | Relaxes   | Relaxes   |
| D | Relaxes   | Contracts |

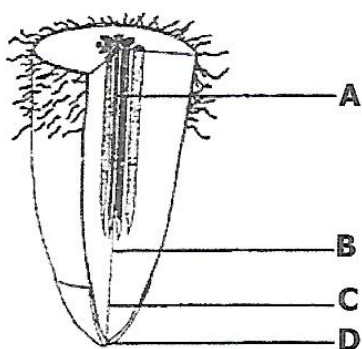
- 24 The diagram shows potted seedlings placed under a cardboard box with a window cut on one side so that light reaches the shoot from one direction.



What type of tropism is being demonstrated in the diagram?

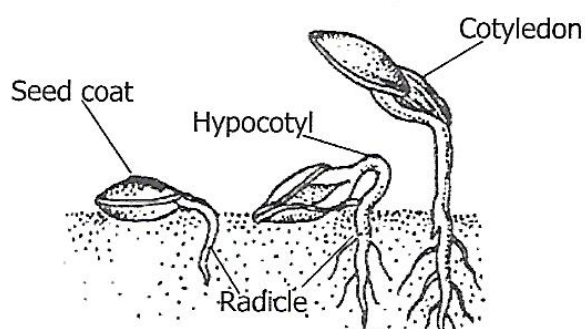
- A Negative geotropism
- B Negative phototropism
- C Positive geotropism
- D Positive phototropism

- 25 The diagram shows a longitudinal section of a root tip.



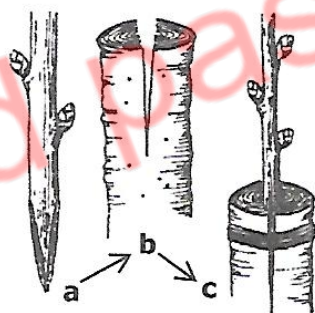
Where does secondary growth occur?

- 26 The diagram shows a germinating seed.



The seedling shown in the diagram shows ...

- A epigeal germination which occurs in dicotyledonous plants.
  - B epigeal germination which occurs in monocotyledonous plants.
  - C hypogeal germination which occurs in dicotyledonous plants.
  - D hypogeal germination which occurs in monocotyledonous plants.
- 27 Which of the following is **not** the importance of fungi?
- A Crop rotation
  - B Nutrient recycling
  - C Production of alcohol
  - D Source of food
- 28 The diagram shows one of the artificial vegetative reproduction methods.

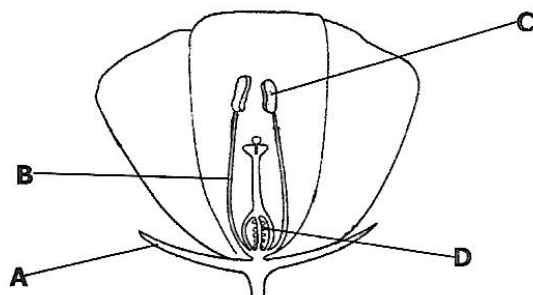


What method of artificial propagation is shown?

- A Budding
- B Cutting
- C Grafting
- D Layering

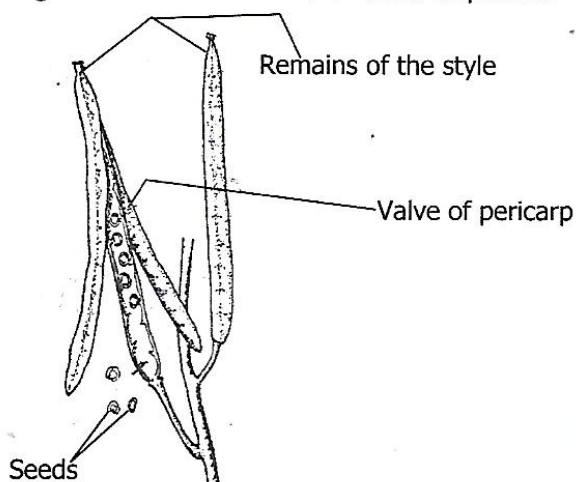


- 29 The diagram shows a section of an insect pollinated flower.



Which structure produces pollen grains?

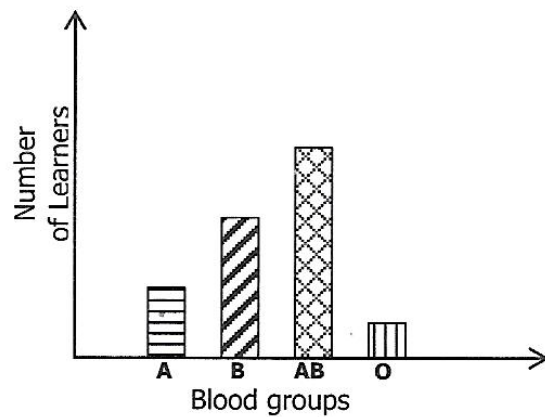
- 30 The diagram shows a method of seed dispersal.



What type of seed dispersal is demonstrated in the diagram?

- A Animal
  - B Self mechanism
  - C Water
  - D Wind
- 31 Which of the following statements is true about reproduction in frogs?
- A External fertilization occurs
  - B Few eggs are laid
  - C Internal fertilization occurs
  - D It is asexual reproduction
- 32 Which part of the reproductive system in humans is cut and tied during vasectomy?
- A Fallopian tube
  - B Sperm duct
  - C Ureter
  - D Urethra

- 33** The following bar graph shows different numbers of learners and their blood groups in a Grade 12 class.

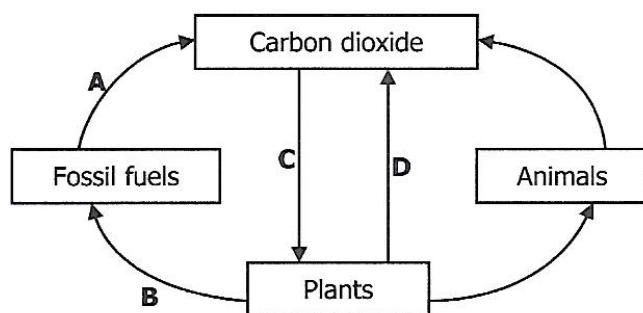


What type of genetic variation is shown in the graph?

- A** Both continuous and discontinuous variation
  - B** Continuous variation
  - C** Discontinuous variation
  - D** Neither continuous nor discontinuous variation
- 34** At which stage of cell division does replication of DNA occur?
- A** Anaphase
  - B** Interphase
  - C** Metaphase
  - D** Telophase
- 35** What are the possible blood groups of the offspring of parents who have blood group **A** and **B**?
- A** AB only
  - B** A and B
  - C** A, B and AB
  - D** A, B, AB and O
- 36** The following is a food chain.  
 Rape → Grasshopper → Bird → Snake  
 Which trophic level is occupied by the grasshopper?
- A** Primary consumer
  - B** Primary producer
  - C** Secondary consumer
  - D** Tertiary consumer
- 37** Which of the following factors increases the size of a population?
- A** Death
  - B** Disease
  - C** Emigration
  - D** Immigration

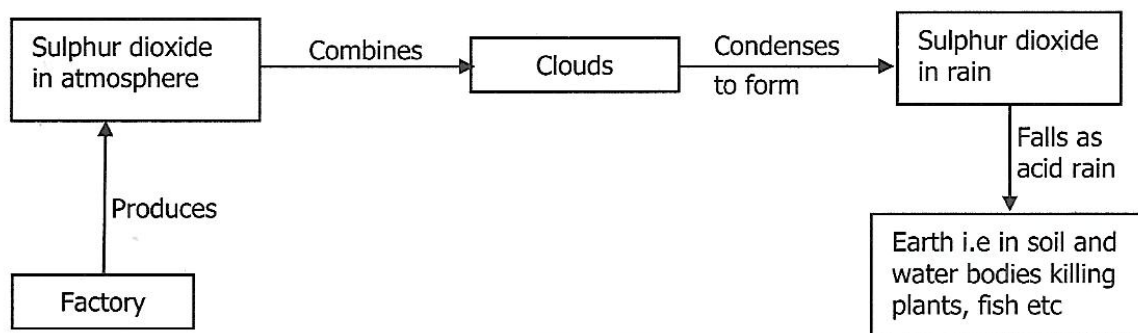


38 The diagram shows a carbon cycle.



Which letter represents combustion?

39 The following diagram illustrates the formation of acid rain due to sulphur dioxide pollution in the atmosphere.



What measures should be put in place to prevent this type of pollution?

- A Deforestation
- B Global warming
- C Soil conservation
- D Using filters in chimneys

40 Which of the following is **not** an adaptive feature which enables insects to elude predators?

- A Compound eyes
- B Larval stage
- C Mouth parts
- D Wings